Blow Lamp

Category: **Welding & Soldering** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: Blow Lamp

What it is used for

A blow lamp uses a controlled flame to heat materials for suitable workshop tasks such as soldering, brazing, or local heating, depending on the model and fuel. Use the specified fuel and ignition procedure, maintain ventilation, and control the flame carefully to prevent burns, fires, explosions, and harmful fumes.

Main parts

- Fuel tank or gas cylinder connection
- Burner nozzle and mixing tube
- Fuel valve or flow regulator
- Ignition system or approved lighting method
- Handle and heat-resistant grip
- Stand, flame adjustment control, and protective components

Before you start

1. Confirm the blow lamp type, approved fuel, and intended heating task.
2. Inspect the fuel container, hose, valve, regulator, seals, burner, and connections for damage or leakage.
3. Check that the nozzle is clean and the correct burner assembly is fitted.
4. Work in a well-ventilated, fire-safe area away from combustible materials and flammable vapours.
5. Secure the workpiece on a stable, heat-resistant surface and protect nearby components from heat.
6. Wear safety glasses or a face shield, heat-resistant gloves, suitable clothing, and other PPE required for the task.

How to use

1. Ensure the fuel valve is closed before connecting or inspecting the fuel supply.
2. Connect the specified fuel source according to the manufacturer's instructions and check for leaks using an approved method.
3. Position the blow lamp on its stable stand or hold it in the designed grip, with the nozzle pointed away from people.
4. Open the fuel valve gradually using the specified startup procedure.
5. Ignite the burner with the approved ignition device or method; never use an improvised ignition source.

6. Adjust the flame to the required stable setting and check for abnormal flame colour, noise, or instability.
7. Direct the flame only at the intended work area and move it steadily to distribute heat as required.
8. Keep the fuel container and hose away from the flame, hot workpiece, sparks, and sharp edges.
9. Monitor the workpiece temperature and surrounding area throughout heating.
10. If the flame goes out, close the fuel valve and follow the manufacturer's relighting procedure after allowing any accumulated gas to disperse safely.
11. When finished, close the fuel valve fully and allow the burner and workpiece to cool in a safe location.
12. Disconnect and store the fuel source only as directed, inspect for leaks, and keep the equipment away from heat and ignition sources.

⚠ Safety

- Only trained users should operate a blow lamp; follow the fuel supplier's and manufacturer's instructions.
- Never use a leaking, damaged, or incorrectly assembled blow lamp.
- Do not heat sealed containers, pressurised vessels, or unknown materials that may release toxic fumes.
- Keep a suitable fire extinguisher nearby and never leave a lit flame unattended.
- Do not use the lamp near flammable liquids, combustible dust, or oxygen-enriched environments.
- Never point the flame toward people, gas cylinders, fuel hoses, or electrical cables.

Care & storage

- Allow the equipment to cool completely before cleaning or storing it.
- Clean the burner nozzle using only the approved method and tools.
- Inspect valves, seals, hoses, and connectors regularly for wear or leakage.
- Store fuel cylinders or containers upright and according to the supplier's safety instructions.
- Remove damaged equipment from service and arrange qualified repair; do not modify the burner or fuel system.

Common mistakes to avoid

- Using the wrong fuel or ignoring a suspected leak.
- Lighting the burner with an unapproved method or opening the valve too quickly.
- Leaving the flame unattended or directing heat toward combustible materials.
- Attempting to relight immediately after flame failure without safely closing the fuel and following the restart procedure.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.